

Visualizing Air Pollution - Pre-reading and Reflection

- A) What is the main cause of air pollution?
- Air pollution is primarily caused by the burning of fossil fuels such as natural gas, which lets off emissions whether it comes from a refinery, power plant, or motorized vehicle. ([IOAir](#))
- B) Why does LA have unhealthy air quality?
- LA has a very large population; The more people you have in one area, the bigger the collective carbon footprint and impact of emissions will be. The area is also vulnerable to wildfires which fill the air with pollutants. ([IOAir](#))
- C) What health problems are caused by air pollution?
- It depends on the pollutant. The [IOAir](#) article goes into detail about two of them; PM2.5 and Ozone. It says “Exposure to PM2.5 has been linked to health effects such as heart disease, respiratory illness, and premature death.”, and “ozone can cause health effects ranging from respiratory infections and inflammation to premature death.” Overall, air pollution leads to us inhaling large amounts of harmful molecules or particles which can mess with our respiratory and immune systems.
- D) Has air quality improved in Los Angeles over the past 5 years?
- According to our [IOAir](#) article, air quality in Los Angeles has increased a lot in the past 30 years, since an amendment was made to the Clean Air Act in 1990. Barely fitting in our 5 year window, Los Angeles experienced a stretch of very good air quality following the lockdown in spring of 2020. As for the years since, things have not been terrible, but LA frequently tops the charts in terms of ozone polluted US cities ([American Lung Association](#))
- E) What is the impact of COVID-19 to air quality?
- Covid-19 caused a lockdown which would majorly improve the air quality of LA for a brief time. The [IOAir](#) article tells us that the air quality met the World Health Organization target, albeit for roughly 3 weeks. This is likely because manufacturing businesses were shut down, less fuels were burned, and less people were using their cars to go anywhere.
- F) What is your neighborhood air quality?
- Air quality for my neighborhood (In Walnut Creek, CA) is decent, swaying between the categories of “good” and “moderate”, especially in the event of a nearby wildfire, one of the largest causes of air pollution for our area. ([IOAir](#))
- G) How could we reduce air pollution in Los Angeles?
- The article on the [IOAir](#) website recommends that residents of Los Angeles switch to using electric vehicles in order to reduce air pollution. Otherwise, supporting positive environmental policies is a great way to crack down on business practices that are unhealthy for air pollution.
- H) What are the major air pollutants?
- For Los Angeles, the major air pollutants are PM2.5 (large airborne particulate matter) and Ozone (coming from vehicular exhaust). Carbon monoxide and nitrogen dioxide are also common air pollutants. All of these pollutants are made up of simple atoms like hydrogen, oxygen, carbon, and nitrogen, which are plentiful even in a healthy atmosphere. Burning of fuels creates an environment where these harmful pollutants can be generated en masse and dispersed into the air. ([IOAir](#))